21 Probability Density Functions of Continuous Random Variables

- The continuous analog of a probability mass function (pmf) is a *probability density function (pdf)*.
- However, while pmfs and pdfs play analogous roles, they are different in one fundamental way; namely, a pmf outputs probabilities directly, while a pdf does not.
- For a continuous random variable X with pdf f_X, the probability that X takes a value in the interval $[a, b]$ is the *area under the pdf over the region* $[a, b]$.
- The **probability density function (pdf)** (a.k.a. density) of a *continuous* RV X is the function f_X which satisfies

$$\mathrm{P}(a \le X \le b) = \int_a^b f_X(x)dx, \qquad \text{for any } -\infty \le a \le b \le \infty \tag{21.1}$$

- The axioms of probability imply that a valid pdf must satisfy

$$f_X(x) \ge 0 \qquad \text{for all } x, \tag{21.2}$$

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{21.3}$$

- A pdf is 0 outside the range of possible values. Given a specific pdf, the generic bounds like $(-\infty, \infty)$ in integrals should be replaced by the range of possible values.

Example 21.1. Recall Example 20.6. Based on earlier examples, it seems reasonable to assume that the pdf f_X is linear as a function of possible x values, with an intercept of 0 and positive slope; that is,

$$f_X(x) = \begin{cases} cx, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Compute the value of c. Note: this involves more than just reading off a value from a plot. What is the key principle that will determine the value of c?

2. Use the pdf (and geometry) to compute $\mathrm{P}(X < 15)$.

3. Use the pdf (and geometry) to compute $\mathrm{P}(X > 45)$.

4. Use the pdf (and geometry) to find an expression for the cdf of X. (Compare to the cdf from Example 20.6.)

- If X is a continuous random variable with pdf f_X, its cdf F_X satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(u)du$$

- Roughly, the cdf is the integral of the cdf

Example 21.2. Continuing Example 21.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

The corresponding cdf satisfies $F_X(x) = (x/60)^2, 0 < x < 60$.

1. Compute and interpret $\mathrm{P}(X = 15)$ and $\mathrm{P}(X = 45)$.

2. Compute the probability that X, truncated to the nearest minute, is 15 minutes after noon; that is, compute $\mathrm{P}(15 \le X < 16)$. How does this probability compare to $f(15)$?

3. Compute the probability that X, truncated to the nearest minute, is 45 minutes after noon; that is, compute $\mathrm{P}(45 \le X < 46)$. How does this probability compare to $f(45)$?

4. Compute and interpret the ratio of the probabilities from the two previous parts. How does this ratio compare to $f(45)/f(15)$?

- The probability that a continuous random variable X equals any particular value is 0. That is, if X is continuous then $\mathrm{P}(X = x) = 0$ for all x.
- In practical applications involving continuous random variables, "equal to" really means "close to", and "close to" probabilities correspond to intervals which can have positive probability.
- The density $f_X(x)$ at value x is *not* a probability. Rather the height of the pdf $f_X(x)$ is related to $\mathrm{P}(X \text{ is close to } x)$ in the following sense.
- Let ϵ be a small number in the measurement units of X. Then

$$\mathrm{P}(x - 0.5\epsilon < X < x + 0.5\epsilon) \approx f_X(x)\epsilon$$

  For example, $\epsilon = 0.01$ represents "close to" as rounding to two decimal places.
- In other words,

$$f_x(x) \approx \frac{\mathrm{P}(x - 0.5\epsilon < X < x + 0.5\epsilon)}{\epsilon} = \frac{F_X(x + 0.5\epsilon) - F_X(x - 0.5\epsilon)}{\epsilon}$$

- Taking the limit as $\epsilon \to 0$, we see that the pdf is the derivative of the cdf

$$f_X(x) = \frac{d}{dx} F_X(x)$$

- For any two possible values x_1 and x_2

$$\frac{\mathrm{P}(X \text{ is close to } x_1)}{\mathrm{P}(X \text{ is close to } x_2)} \approx \frac{f_X(x_1)}{f_X(x_2)}$$

  regardless of how "close to" is defined.

Example 21.3. Continuing Example 20.5. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \qquad x \geq 0$$

1. Sketch the pdf of X. What does this tell you about waiting times?

2. Without doing any integration, approximate the probability that X rounded to the nearest second is 30 minutes.

3. Without doing any integration determine how much more likely that X rounded to the nearest second is to be 30 than 60 minutes.

4. Compute and interpret $\mathrm{P}(X < 30)$.

5. Compute and interpret $\mathrm{P}(X > 360)$.

6. Find the cdf of X.

7. Find the 25th percentile of X.

8. Find the quantile function of X.

9. Sketch a spinner for simulating values of X.